

Classroom Assignment <1>: Univariate, Bivariate, and Multivariate Analysis

Learning Objective:

Understand **univariate, bivariate, and multivariate analysis** using Pandas DataFrames loaded from CSV.

Expected Completion Time:

Best Case: 25 minutes

Average Case: 35 minutes

Assignment Details:

TestCase	Module	Status	Duration	Defects
TC1	Login	Passed	12	0
TC2	Login	Failed	15	2
TC3	Payment	Passed	20	1
TC4	Payment	Failed	18	3
TC5	Reports	Passed	25	0

Write a Python script **pandas_univariate_bivariate.py** that:

- Reads the CSV into a DataFrame.

Univariate Analysis

- Show distribution of one column: `Duration`.
- Print mean, median, std of `Duration`.

Bivariate Analysis

- Compare `Status` vs `Duration`. (e.g., average duration by status).
- `df.groupby("Status")["Duration"].mean()`

Multivariate Analysis

- Compare `Module`, `Status`, and `Defects`.
- Example: `df.groupby(["Module", "Status"])["Defects"].sum()`

Hints to Solve:

- Use `pd.read_csv("test_results.csv")`.
- Use `.describe()` for univariate summary.
- Use `.groupby()` for bivariate/multivariate.

Expected Outcome: Understand how to analyze **one variable, two variables, and multiple variables** from the same dataset.